



RTL Design Sherpa

Binary BCH Codec Micro-Architecture Specification 0.1 (draft)

October 3, 2026

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List of Waveforms

No waveforms in this document.

1 Bch Mas Index

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2 Document Information

2.1 Binary BCH Codec Micro-Architecture Specification

Property	Value
Document Title	Binary BCH Codec Micro-Architecture Specification
Version	0.1 (draft)
Date	October 3, 2026
Status	Draft — no RTL exists; every structural choice is either carried from the PRD or open and tied to a PRD decision ID
Classification	Open Source - MIT License

2.2 Revision History

Version	Date	Author	Description
0.1	2026-10-03	RTL Design Sherpa	First draft, written before any RTL exists. Carries the HAS architecture one level down: per-block signal-level intent, FSM policy, timing, and signal-contract references. Numbers are analytic placeholders derived from parameters; none are measured.

2.3 Document Purpose

This Micro-Architecture Specification (MAS) is the implementation-level view of the Binary BCH codec — the HAS (`./bch_has/`) taken one level down. It is the document an RTL author reads before writing SystemVerilog and a verification engineer reads before writing assertions.

It covers:

- each functional block's internal structure, signals, and parameters
- the FSM policy for every block: where a state machine is forbidden, where one minimal control FSM is permitted, and what replaces FSMs elsewhere
- cycle-level timing and throughput placeholders tied to the open PRD decisions
- the signal-contract posture that will be checked by the generated kmap workbook

2.4 Intended Audience

- RTL implementers writing the encoder, decoder, and GF-datapath blocks
- Verification engineers checking the golden-model equivalence and the formal properties in `formal`
- Architects reviewing whether a candidate D6 throughput or D11 solver choice fits a consumer's area/latency target

2.5 Related Documents

Table 0.1: Related documents

Document	Location	Content
Product Requirements	<code>../.. /PRD.md</code>	the decision table (D1-D12) and the candidate profiles
Hardware Architecture Specification	<code>../bch_has/bch_has_index.md</code>	high-level block diagram, data flow, solver options, interfaces, parameters
Signal-contract workbook	<code>../.. /bch_signal_contracts.xlsx</code>	generated contract sheets and kmap decision tables
Signal-contract generator	<code>../.. /gen_bch_signal_contracts_kmaps.py</code>	Python generator for the workbook
References	<code>../.. /References/README.md</code>	the standards and papers, with source and licence
CLAUDE.md	<code>../.. /CLAUDE.md</code>	area facts for a session working here
Handbook	<code>vault/handbook/design/</code>	streaming-no-fsm, minimal-fsm, signal-contracts-and-kmaps, valid-ready contracts

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3 Methodology

3.1 HAS to MAS relationship

The Hardware Architecture Specification (`../bch_has/`) records what the BCH codec is, where it sits, what it presents at its boundaries, and what parameters configure it. This Micro-Architecture Specification carries that architecture one level down: per-block signal tables, cycle behavior, FSM policy, and the signal-contract references that will become the formal pre-RTL design review.

The rule in this repo is one source per fact. The MAS does not restate the PRD decision table or the HAS block diagram; it cross-references them and adds the implementation detail that belongs at this level. Parameters and throughput numbers that are not yet fixed are marked **TBD** and tied to the PRD decision ID that fixes them, following the convention in `../bch_has/ch06_integration/02_parameters.md`.

3.2 FSM policy

The repo's datapath FSM rule, taken from the design handbook, is:

No FSM in the per-beat data path. Not a minimal one, not a two-state one. FSMs belong to control: descriptor lifecycle, schedulers, init sequencers, error recovery. If a state machine can observe a data beat, it is in the wrong place. — vault/handbook/design/streaming-no-fsm.md

What replaces a datapath FSM in this block:

Instead of a state that...	Write
counts beats, or waits N cycles	a counter plus a comparator
remembers “I am mid-burst”	a <code>r_valid</code> flag on the pipeline stage
serializes because a resource is shared	an arbiter and a qualifier, not a turn-taking machine
gates until a condition holds	a qualifier ANDed into the handshake

Where a state machine is genuinely justified, the handbook requires:

Fewest states that carry the real distinctions. A state that only waits one fixed cycle, or that differs from a sibling only by a datapath value, is a register, not a state. Merge states whose outputs and successors are identical. Idiom: `typedef enum logic [N-1:0] {...} state_t; + two-process form - registered r_state, combinational w_next_state with a default-hold first line and a default arm. One FSM per module; nested/communicating FSMs are a smell that the block wants splitting. Outputs: prefer registered (Moore) at module boundaries; combinational decode of r_state internally is fine but belongs in the K-maps. — vault/handbook/design/minimal-fsm.md`

Every block page in this chapter states its FSM policy explicitly and cites this page. Blocks that are purely datapath carry no FSM; the one block that sequences decoder stages carries exactly one minimal control FSM.

3.3 Prove-by-design / kmap philosophy

The signal-contract and kmap discipline in this repo treats the act of mirroring RTL expressions in a computed workbook as a design review in itself:

The act of mirroring expressions IS a design review. The stream workbook's first generation surfaced six findings the test suite had not...
— vault/handbook/design/signal-contracts-and-kmaps.md

For the BCH MAS, that posture is adopted pre-RTL: the contracts in `bch_signal_contracts.xlsx` cite the MAS pages where the intended expressions are written verbatim. When RTL lands, those citations are re-pointed at the relevant `.sv` file and line. The workbook is generated by `gen_bch_signal_contracts_kmaps.py`; it is never hand-edited.

3.4 Carrying open PRD decisions

Every undecided item in the PRD is carried as **TBD** with the decision ID that resolves it. Examples in this document:

Parameter / choice	PRD decision
field dimension m	D1
correctable bits t	D2
block length N_BITS	D3
encoder/decoder build mix	D4
erasure support	D5
throughput architecture and cycle counts	D6
bits per beat $BITS_PER_BEAT$	D9
key-equation solver algorithm	D11

No numeric value is invented. Cycle-count formulas are stated in terms of the parameters above (for example, N_BITS cycles per syndrome pass in the serial case), and throughput trade-offs are placeholders tied to D6.

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4 Block Inventory

The MAS documents the target blocks below. Each page follows the FUB-page skeleton from the pumice MAS: header metadata, purpose, parameters, interface, microarch internals, timing, and notes. The $GF(2^m)$ primitives are imported from the reed-solomon component per PRD D7 and are not re-documented here.

Table 1.1: MAS block inventory

Block	RTL target	Role	MAS page
bch_encoder_core	rtl/ bch_encoder_core .sv	Systematic encoder: bit serialiser, GF LFSR parity generator, parity mux, bit packer	01_encoder.md
bch_syndrome_unit	rtl/ bch_syndrome_unit .sv	Computes the t odd syndromes $S_1 \dots S_{2t-1}$ over the serial bit stream using t parallel GF MAC lanes	02_syndrome_unit .md
bch_key_equation_solver	rtl/ bch_key_equation_solver.sv	Inverts the key equation from odd syndromes to error-locator $\Lambda(x)$; D11 selects riBM, Euclid, or step-by- step/PGZ	03_key_equation_solver.md
bch_chien_search	rtl/ bch_chien_search .sv	Evaluates $\Lambda(x)$ at all N_BITS bit positions, flags roots, flips bits; no Forney stage	04_chien_search. md
bch_decoder_core	rtl/ bch_decoder_core	Block buffer + stage sequencer	05_decoder_core.

Block	RTL target	Role	MAS page
	.sv	between syndrome, solver, Chien, and corrector; exactly one minimal control FSM	md

The GF layer is imported, not duplicated:

Table 1.2: Imported GF primitives

Primitive	Source in projects/components/ecc-ip/reed-solomon/rtl/fub/gf/	Used by
gf_pkg	gf_pkg.sv	all blocks (tables, constants, generator polynomial)
gf_mul	gf_mul.sv	syndrome unit, key-equation solver, Chien search
gf_mul_const	gf_mul_const.sv	LFSR encoder taps, Chien Horner cells
gf_inv	gf_inv.sv	Euclidean solver only (if D11 selects EUCLID)

The adapter blocks (axis_adapter, axi4_job_adapter, register block) are specified in ../bch_has/ch04_interfaces/ and are not taken to the signal level in this MAS.

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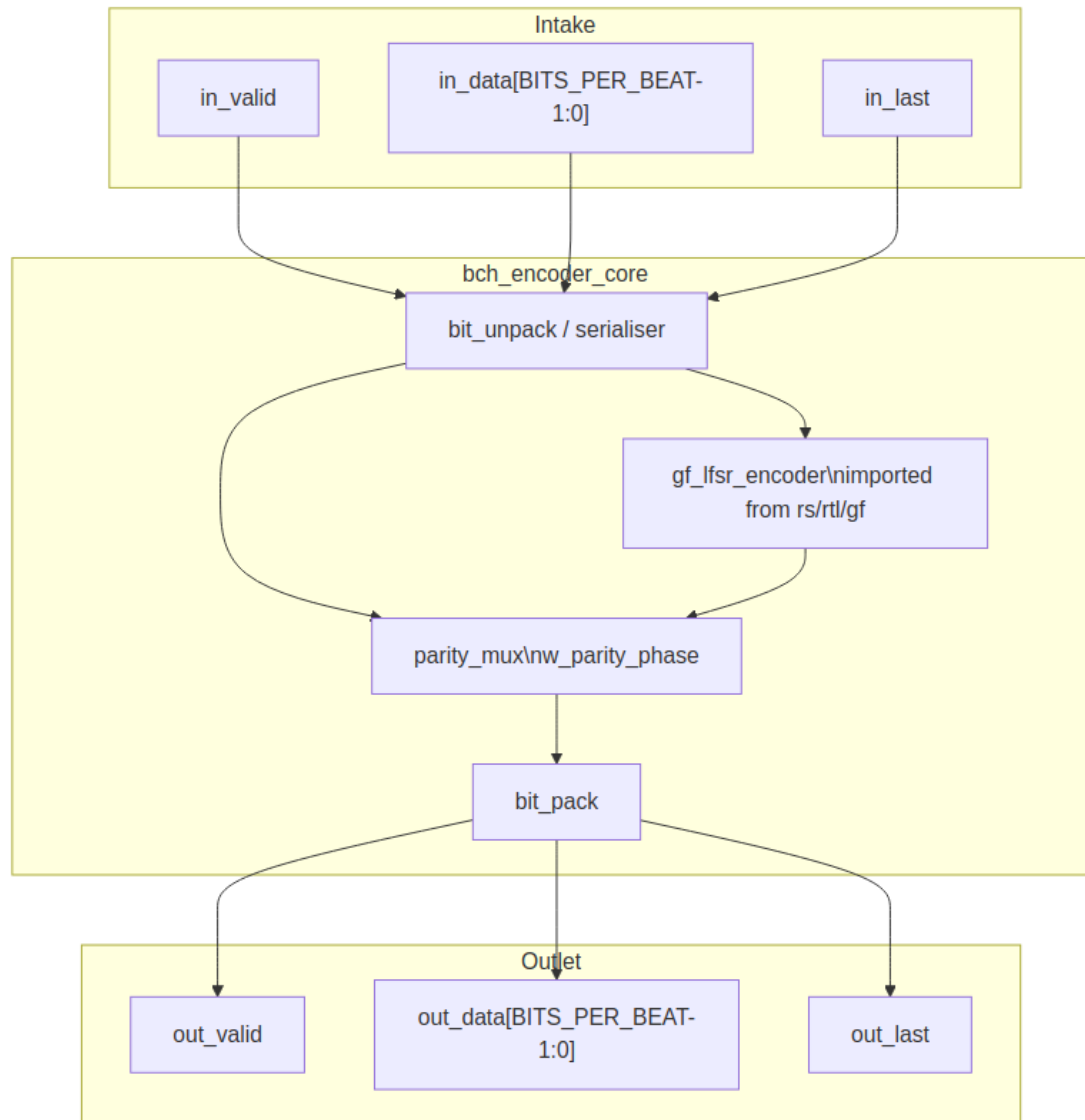
5 Encoder Core (bch_encoder_core)

Module: bch_encoder_core.sv **Location:** rtl/macro/ **Category:** streaming datapath **Parent:** bch_top / standalone encoder wrapper **Status:** landed — rtl/macro/bch_encoder_core.sv, gate DV green on the three standing profiles

5.1 Purpose

bch_encoder_core systematically encodes a K_BITS-bit data block into an N_BITS-bit BCH codeword. Data bits pass through unchanged; the block then appends the N_BITS - K_BITS parity bits produced by a GF(2^m) LFSR whose taps come from the generator polynomial computed by gf_pkg. The output stream is longer than the input stream, so the encoder is not a pure back-to-back pipeline: it holds in_ready low while parity drains.

5.1.1 Figure 2.1: Encoder core block diagram



Encoder core block diagram

5.2 Parameters

Table 2.1: Encoder core parameters

Parameter	Type	Range	Default	Meaning	PRD
FIELD_DIM	int	3..16	TBD	m; field is GF(2 ^m)	D1
T_BITS	int	1..(2 ^m -1)/2	TBD	correctable bit errors	D2
N_BITS	int	2t+1 .. 2 ^m -1	TBD	codeword length in bits	D3
BITS_PER_BEAT	int	1 .. K_BITS	TBD	bits per valid/ready beat	D9

Derived: $K_BITS = N_BITS - DEGREE_G$, where $DEGREE_G$ is the degree of the generator polynomial computed by `gf_pkg` from m , t , and the profile parameters. $DEGREE_G$ is at most $m * T_BITS$ and equals $N_BITS - K_BITS$.

5.3 Interface

Table 2.2: Encoder core ports

Signal	Direction	Width	Description
aclk	in	1	one clock for the whole core
aresetn	in	1	active-low synchronous reset
in_valid	in	1	a beat of data bits is offered
in_ready	out	1	the core takes it this cycle
in_data	in	BITS_PER_BEAT	data bits, bit 0 in the low bit; partial final beat uses in_keep
in_keep	in	BITS_PER_BEAT	present-bit mask, low-aligned; partial only on the block's final beat
in_last	in	1	this beat carries the K_BITS-th data bit
out_valid	out	1	a beat of coded bits is offered
out_ready	in	1	the consumer takes it
out_data	out	BITS_PER_BEAT	data bits while ! w_parity_phase, parity bits while w_parity_phase
out_keep	out	BITS_PER_BEAT	present-bit mask; partial at the end of data and at the end of parity
out_last	out	1	this beat carries

Signal	Direction	Width	Description
frame_err	out	1	the N_BITS-th coded bit pulse: a block ended with other than K_BITS data bits

5.4 Microarchitecture internals

5.4.1 Bit serialiser / unpack

The input beat `in_data` is serialised beat-by-beat into a single bit stream. With `BITS_PER_BEAT = 1` this is a pass-through. With `BITS_PER_BEAT > 1` a small barrel or counter-driven mux presents one valid bit per cycle to the LFSR. The `in_keep` mask qualifies the final beat so that only real data bits enter the LFSR.

5.4.2 GF(2^m) LFSR parity generator

The parity generator is imported from the reed-solomon component per PRD D7:

`gf_lfsr_encoder` in

`projects/components/ecc-ip/reed-solomon/rtl/fub/gf/gf_lfsr_encoder.sv`. It is a 2^m * `T_BITS`-stage shift register over GF(2^m) with constant multipliers on the taps derived from $g(x)$. The encoder clears at block start, consumes the `K_BITS` data bits serially, and then holds the parity symbols.

The binary BCH use is identical to the RS encoder shape; the only difference is that the input is bits, not symbols, so the LFSR advances once per bit.

5.4.3 Parity mux

The output mux selects data bits while the block is in the data phase and parity bits once the data phase completes:

```
w_parity_phase = (r_bit_count >= K_BITS) && w_beat_is_parity
```

`w_beat_is_parity` is true whenever the current output position is inside the parity region; it is derived from the same bit counter:

```
w_beat_is_parity = (r_bit_count + BITS_PER_BEAT - 1) >= K_BITS
```

The mux output is `out_data = w_parity_phase ? w_parity_bits : w_data_bits`.

5.4.4 Bit packer

The serial output stream is reassembled into `BITS_PER_BEAT`-bit output beats. `out_keep` is full except on the final data beat and the final parity beat, where it marks the partial beat. `out_last` raises on the beat that carries the `N_BITS`-th coded bit:

```
w_enc_out_last = w_parity_phase && (r_parity_count == N_BITS - K_BITS - 1)
```

5.5 FSM policy

This block carries **no FSM** (per vault/handbook/design/streaming-no-fsm.md). The sequencing that another design might put in a state machine is replaced by:

- `r_bit_count`: counts input/output bit position in the block, 0 .. `N_BITS-1`
- `r_parity_count`: counts parity bits emitted, 0 .. `N_BITS-K_BITS-1`
- `r_valid` pipeline flags on the packer output stage
- `w_parity_phase`: combinational qualifier selecting parity vs data
- `w_last`: combinational qualifier marking the final output beat

`in_ready` is combinational `!r_valid || m_ready` at the input skid, ANDed with the qualifier that the block is not currently draining parity.

5.6 Timing

- One input beat is accepted per cycle while `in_ready` is high.
- The data phase is $\text{ceil}(K_BITS / BITS_PER_BEAT)$ beats.
- The parity phase is $\text{ceil}((N_BITS - K_BITS) / BITS_PER_BEAT)$ beats.
- `in_ready` is held low during the parity phase; back-to-back blocks need an upstream FIFO sized for at least $N_BITS - K_BITS$ bits.
- The exact blocks-per-cycle numbers are placeholders tied to PRD D6 and D9.

5.7 Notes

- **Wrong-length block handling:** If `in_last` arrives before or after the `K_BITS`-th data bit, `frame_err` pulses and the block is still encoded as given (systematic data pass-through with parity appended to whatever data arrived). This is the encoder-side analog of the decoder's `out_status_frame_err` (R1/R2):

`w_frame_err = in_last && (r_bit_count != K_BITS - 1)`

- **OFF-state testing convention:** For tests that need to verify the core tolerates idle cycles, the LFSR clears on `in_last` acceptance and the output pipeline stalls cleanly when `out_ready` is low.
- The `gf_lfsr_encoder` instance is configured by the same `FIELD_DIM`, `T_BITS`, `N_BITS`, `PRIM_POLY`, and `FIRST_ROOT` parameters as the rest of the codec; its generator polynomial comes from `gf_pkg` (PRD D3/D8).

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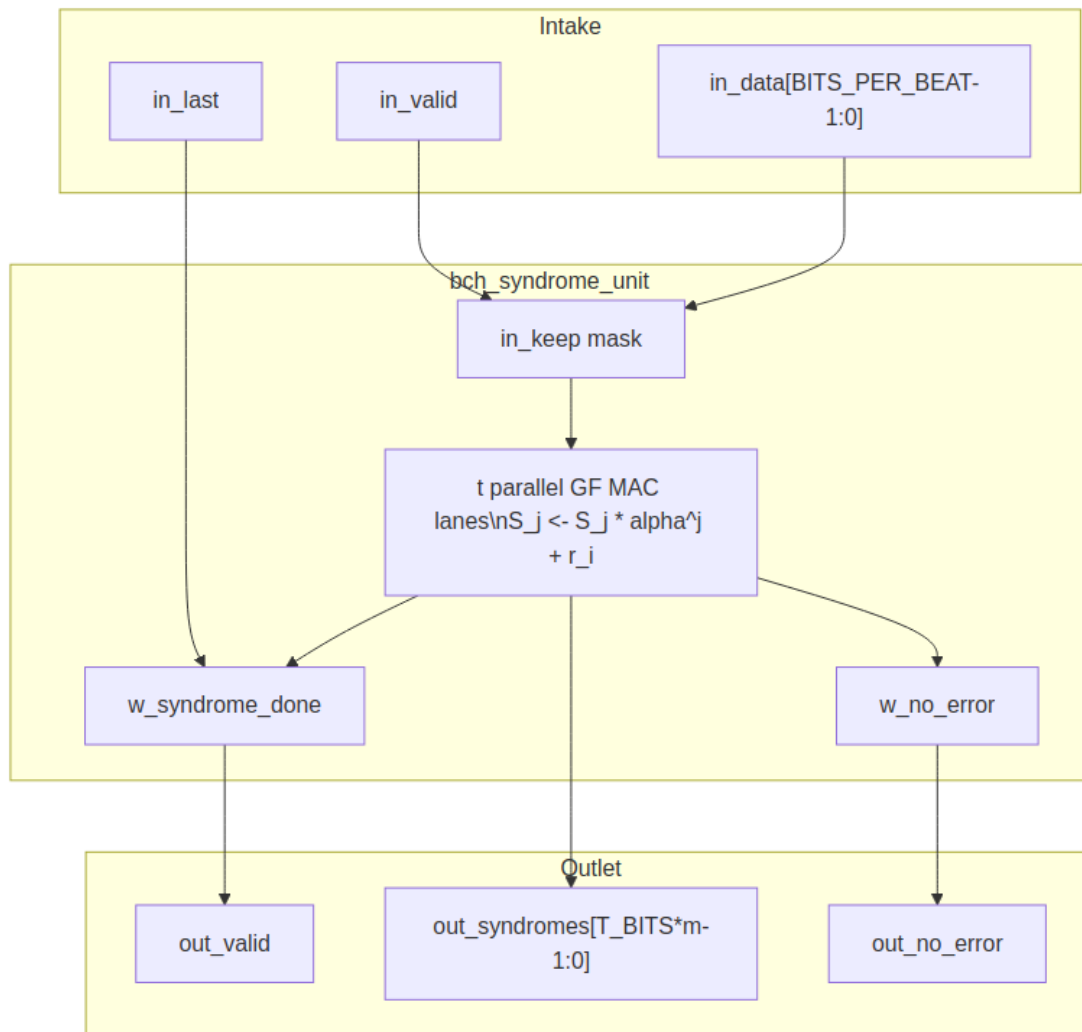
6 Syndrome Unit (bch_syndrome_unit)

Module: bch_syndrome_unit.sv **Location:** rtl/fub/ **Category:** streaming datapath **Parent:** bch_decoder_core **Status:** landed — rtl/fub/bch_syndrome_unit.sv, gate DV green on the three standing profiles

6.1 Purpose

bch_syndrome_unit computes the t odd syndromes $S_1, S_3, \dots, S_{2t-1}$ of the received BCH codeword as the bits arrive. The even syndromes are not computed directly; they are recovered from the odd ones through the binary BCH evenness shortcut $S_{2j} = S_j^2$ (PRD §2). The unit contains t parallel $GF(2^m)$ multiply-accumulate lanes, one per odd syndrome.

6.1.1 Figure 2.2: Syndrome unit block diagram



Syndrome unit block diagram

6.2 Parameters

Table 2.3: Syndrome unit parameters

Parameter	Type	Range	Default	Meaning	PRD
FIELD_DIM	int	3..16	TBD	m; field is GF(2^m)	D1
T_BITS	int	$1..(2^m-1)/2$	TBD	correctable bit errors	D2
N_BITS	int	$2t+1 .. 2^m-1$	TBD	codeword length in bits	D3
BITS_PER_BEAT	int	1 .. K_BITS	TBD	bits per valid/ready beat	D9

6.3 Interface

Table 2.4: Syndrome unit ports

Signal	Direction	Width	Description
aclk	in	1	one clock
aresetn	in	1	active-low synchronous reset
in_valid	in	1	received bits are offered
in_ready	out	1	unit takes them this cycle
in_data	in	BITS_PER_BEAT	received bits, low- aligned
in_keep	in	BITS_PER_BEAT	present-bit mask for partial final beat
in_last	in	1	this beat carries the N_BITS-th received bit
out_valid	out	1	t odd syndromes are valid
out_ready	in	1	downstream accepts the syndromes
out_syndromes	out	$T_BITS * m$	$\{S_{2t-1}, \dots,$ $S_3, S_1\}$ packed GF values
out_no_error	out	1	all odd syndromes are zero

6.4 Microarchitecture internals

6.4.1 Odd-syndrome recurrence

For a received polynomial $r(x) = \sum_{i=0}^{N_BITS-1} r_i x^i$, the odd syndromes are:

$$S_j = r(\alpha^j) = \sum_i r_i * (\alpha^j)^i \quad \text{for } j = 1, 3, \dots, 2t-1$$

As bits arrive one at a time, each lane accumulates by Horner's rule:

$$S_j \leftarrow S_j * \alpha^j + r_i$$

The multiplier by the constant α^j is a `gf_mul_const` instance (imported from `projects/components/ecc-ip/reed-solomon/rtl/fub/gf/gf_mul_const.sv`). No per-position power table is stored; the constant α^j is wired once per lane.

With `BITS_PER_BEAT = B > 1`, each lane performs B Horner steps per cycle:

$$S_j \leftarrow (((S_j * \alpha^j) + r_{i0}) * \alpha^j + r_{i1}) * \alpha^j + \dots + r_{iB-1}$$

This is a short feed-forward chain of B constant multiplies and adds.

6.4.2 All-zero bypass qualifier

The no-error bypass is a combinational qualifier, not a state:

$$w_no_error = (S_1 == 0) \&\& (S_3 == 0) \&\& \dots \&\& (S_{2t-1} == 0)$$

When `w_no_error` is true on the cycle `w_syndrome_done` raises, the decoder core skips the key-equation solver and Chien search and drains the block buffer unchanged.

6.4.3 Completion qualifier

The syndromes are complete once the last received bit has been accumulated:

$$w_syndrome_done = (r_bit_count == N_BITS - 1) \&\& in_valid \&\& in_ready$$

`out_valid` is a registered flag set by `w_syndrome_done` and cleared by the handshake with the downstream solver.

6.5 FSM policy

This block carries **no FSM** (per vault/handbook/design/streaming-no-fsm.md). Control is reduced to:

- `r_bit_count`: counts received bits, $0 \dots N_BITS - 1$
- `r_lane_valid`: registered output valid flag
- `w_syndrome_done`: combinational completion qualifier
- `w_no_error`: combinational all-zero qualifier

Each lane is a pure multiply-accumulate datapath; the lane count and the output valid flag replace any state machine.

6.6 Timing

- One received bit advances every lane by one Horner step.
- Serial case ($\text{BITS_PER_BEAT} = 1$): N_BITS cycles per block.
- Parallel-beat case: $\text{ceil}(N_BITS / \text{BITS_PER_BEAT})$ cycles per block, with B chained GF constant multiplies per lane per cycle.
- The exact cycle count is a placeholder tied to PRD D6.

6.7 Notes

- A shortened code ($N_BITS < 2^m - 1$) is handled by treating the missing leading bits as zero: the Horner recurrence naturally accumulates starting from $S_j = 0$, which is equivalent to zero-padding the high-order coefficients.
- The evenness shortcut means the formal proof targets include $S_{2j} == gf_square(S_j)$ for every odd j ; the square in $GF(2^m)$ is a linear bit-permutation and is cheap to verify.
- `out_no_error` is a combinational decode of the syndrome registers; it is stable from `w_syndrome_done` until the output handshake clears it.

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7 Key Equation Solver (bch_key_equation_solver)

Module: bch_key_equation_solver.sv **Location:** rtl/fub/ **Category:** datapath + conditional control FSM **Parent:** bch_decoder_core **Status:** landed —
rtl/fub/bch_key_equation_solver.sv implements “RIBM” (the documented default when no consumer names a small-t profile); KES_ALGO values other than “RIBM” raise an elaboration error rather than silently falling back. PRD D11 stays open for EUCLID / SMALL_T.

7.1 Purpose

`bch_key_equation_solver` inverts the key equation from the t odd syndromes $S_1, S_3, \dots, S_{2t-1}$ to the error-locator polynomial $\Lambda(x)$. The choice of algorithm is PRD D11; this page documents all three candidates and their microarchitectural implications. Nothing outside the solver knows which algorithm produced Λ : the Chien search, block buffer, and corrector are identical in every build.

7.2 Parameters

Table 2.5: Key-equation solver parameters

Parameter	Type	Range	Default	Meaning	PRD
FIELD_DIM	int	3..16	TBD	m; field is GF(2^m)	D1
T_BITS	int	$1..(2^m-1)/2$	TBD	correctable bit errors	D2
KES_ALGO	string	"RIBM", "EUCLID", "SMALL_T"	TBD	solver algorithm	D11

7.3 Interface

Table 2.6: Key-equation solver ports

Signal	Direction	Width	Description
aclk	in	1	one clock
aresetn	in	1	active-low synchronous reset
in_valid	in	1	odd syndromes are valid
in_ready	out	1	solver accepts syndromes
in_syndromes	in	$T_BITS * m$	$\{S_{2t-1}, \dots, S_3, S_1\}$ packed GF values
out_valid	out	1	$\Lambda(x)$ coefficients are valid
out_ready	in	1	downstream accepts Λ
out_lambda	out	$(T_BITS + 1) * m$	locator coefficients $\Lambda_0 \dots \Lambda_{T_BITS}$
out_lambda_degree	out	$\lceil \log_2(T_BITS + 1) \rceil$	actual degree of Λ (clamped to T_BITS when <code>out_more_than_t</code>)
out_more_than_t	out	1	any Λ_i above index T_BITS is nonzero — more than t errors; the decoder core folds it into the R2 verdict

7.4 Candidate algorithms

The trade-off table below is copied from
`../bch_has/ch03_architecture/03_solver_options.md`.

Table 2.7: Key-equation solver candidates

	inversionless BM ("RIBM")	modified Euclidean ("EUCLID")	step-by-step / PGZ ("SMALL_T")
Algorithm	reformulated inversionless Berlekamp-Massey, Sarwate and Shanbhag 2001; seeded with the t odd syndromes	Sugiyama's Euclidean algorithm with cross-multiplication in place of division (Shao et al. 1985)	Massey 1965 step-by-step correction, or the Peterson-Gorenstein-Zierler direct matrix solve for small t
Array	$3t + 1$ processing elements, 2 registers each	four register arrays: R and Q of $2t + 1$ symbols, the shifted locator pair of $2t + 3$	control-heavy but small datapath: one or a few GF multipliers and a syndrome update loop
GF multipliers	2 per element: $6t + 2$	2 per coefficient of each pair: $8t + 8$	$O(t^2)$ logic or a single iterative multiplier
Iterations	exactly $2t$	data-dependent, $t + 1 \dots 2t$	one per located error position, up to t
Critical path per iteration	one multiply and one XOR, no feedback across the array	cross-multiply feeding a degree comparison that drives the swap: longer	depends on implementation; often dominated by syndrome-update MUX tree
Control	discrepancy select and gamma update	two nominal degree counters, the normalise /	position counter, syndrome-test loop, error-

	inversionless BM ("RIBM")	modified Euclidean ("EUCLID")	step-by-step / PGZ ("SMALL_T")
		cross / swap rule	counter update
Readability	dense; the reformulation is not obvious from the textbook	the textbook algorithm, recognisable step by step	the simplest conceptually for very small t
Best fit	general t; the default if no consumer names a small-t profile	a cross-check and a readable alternative	$t \leq 8$ (flash pages often sit here); smallest area at tiny t

No default is selected at MAS v0.1. The choice is PRD D11.

7.5 Microarchitecture internals

7.5.1 "RIBM" and "EUCLID" — fixed-iteration arrays

Both the reformulated inversionless Berlekamp-Massey array and the modified Euclidean array are pure datapath structures. They need:

- an iteration counter, $0 \dots 2 * T_BITS$
- a start pulse from `in_valid` && `in_ready`
- a completion qualifier when the iteration count reaches its target

Neither implementation has a state machine in the data path. The control signals (discrepancy select, gamma update, degree comparison, swap) are combinational or registered qualifiers derived from the iteration counter and the array registers.

7.5.2 "SMALL_T" — conditional minimal control FSM

The step-by-step and PGZ approaches are control-heavy. If D11 selects "SMALL_T", this block is the one place in the decoder datapath where a minimal control FSM is permitted (per [vault/handbook/design/minimal-fsm.md](#)). The target state list is the fewest states that carry real distinctions:

```
IDLE    -> wait for in_valid
RUN     -> execute one iteration / error-position test
TEST    -> evaluate the discrepancy / matrix determinant
DONE    -> out_valid raised, hold until out_ready
```

The IDLE/RUN/TEST/DONE list is conditional on `KES_ALGO == "SMALL_T"` and is marked as such in the RTL. States that only wait one fixed cycle are merged into RUN with a sub-counter.

7.6 FSM policy

- For "RIBM" and "EUCLID": **no FSM** in the data path; only an iteration counter and completion qualifier.
- For "SMALL_T": **one minimal control FSM** is permitted, conditional on D11, with the state list above.

7.7 Timing

- "RIBM": exactly $2 * T_BITS$ iterations.
- "EUCLID": between $T_BITS + 1$ and $2 * T_BITS$ iterations, data-dependent.
- "SMALL_T": up to T_BITS error-position tests, plus setup cycles.
- Exact cycle counts are placeholders tied to PRD D11 and D6.

7.8 Notes

- Two solvers agreeing as a cross-check is the recommended verification posture from the HAS: the golden model runs every candidate on every block and requires identical roots, and the RTL is built with whichever algorithm the first consumer prefers.
- The locator degree `out_lambda_degree` is produced by the solver and is used by the Chien search to determine correctability.
- `Lambda_0` is always 1 for a valid locator; the formal targets include this invariant and the relationship to the odd syndromes.

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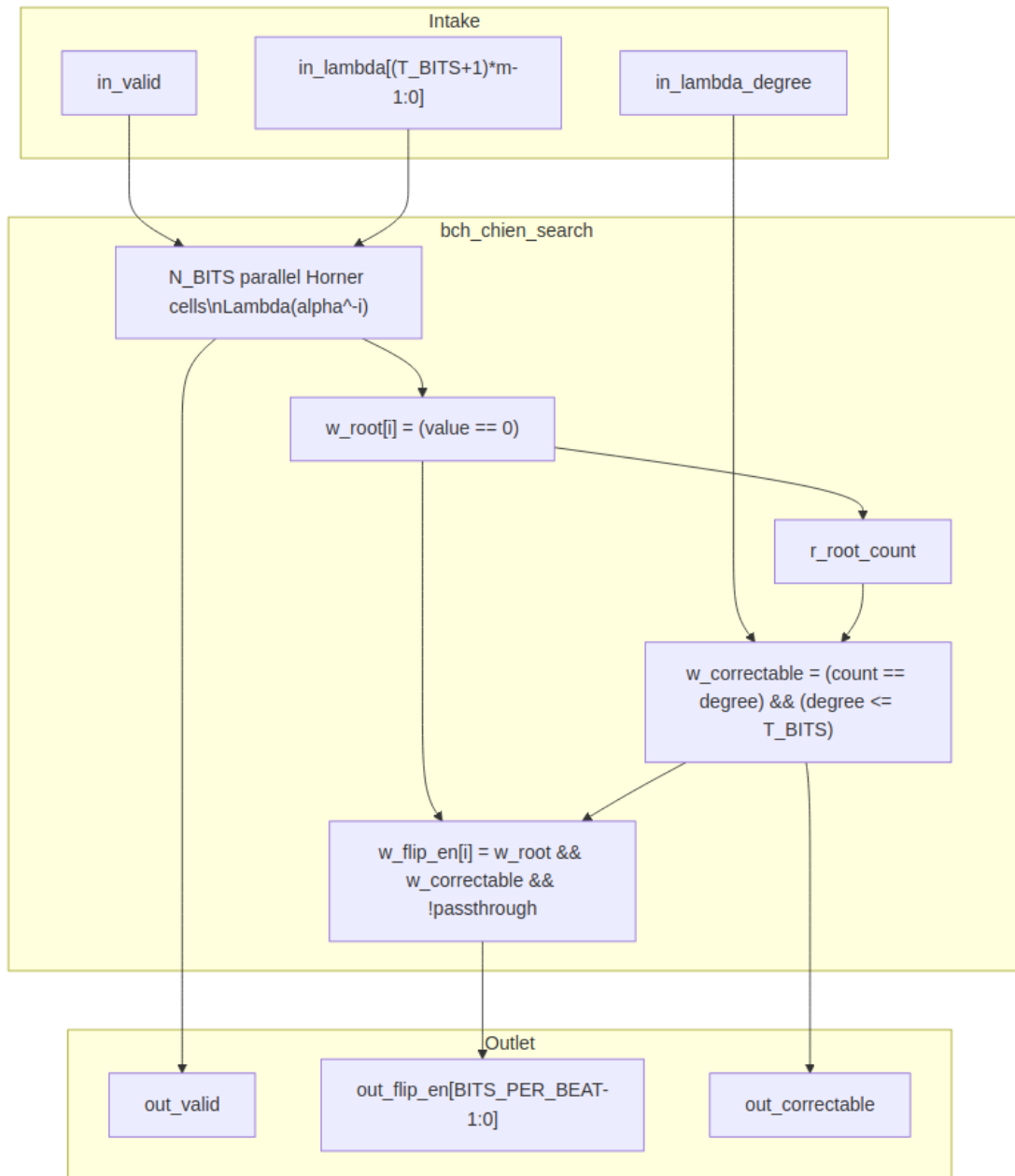
8 Chien Search (bch_chien_search)

Module: bch_chien_search.sv **Location:** rtl/fub/ **Category:** streaming datapath **Parent:** bch_decoder_core **Status:** landed — rtl/fub/bch_chien_search.sv. The flip-enable qualification below is resolved: out_flip_en carries the raw root flags per beat; the w_correctable && !w_release_passthrough qualification is a decoder-core output function (release-on-verdict), applied where the verdict is final, not in this walk.

8.1 Purpose

bch_chien_search evaluates the error-locator polynomial $\Lambda(x)$ at every one of the N_BITS bit positions of the received codeword. Where Λ has a root, the corresponding bit is in error. Because the code is binary, every error value is 1 (PRD §2); the block therefore flips the bit directly and needs no Forney evaluator. The block also feeds the uncorrectability verdict by comparing the root count to the locator degree.

8.1.1 Figure 2.3: Chien search block diagram



Chien search block diagram

8.2 Parameters

Table 2.8: Chien search parameters

Parameter	Type	Range	Default	Meaning	PRD
FIELD_DIM	int	3..16	TBD	m; field is $GF(2^m)$	D1
T_BITS	int	$1..(2^m-1)/2$	TBD	correctable bit errors	D2
N_BITS	int	$2t+1 .. 2^m-1$	TBD	codeword length in bits	D3
BITS_PER_BEAT	int	1 .. K_BITS	TBD	bits per valid/ready beat	D9

8.3 Interface

Table 2.9: Chien search ports

Signal	Direction	Width	Description
aclk	in	1	one clock
aresetn	in	1	active-low synchronous reset
in_valid	in	1	$\text{Lambda}(x)$ coefficients are valid
in_ready	out	1	unit accepts Lambda
in_lambda	in	$(T_BITS + 1) * m$	locator coefficients $\text{Lambda}_0 \dots$ Lambda_{T_BITS}
in_lambda_degree	in	$\lceil \log_2(T_BITS + 1) \rceil$	actual degree of Lambda
out_valid	out	1	a beat of correction flags is offered
out_ready	in	1	downstream accepts correction flags
out_flip_en	out	$BITS_PER_BEAT$	one-hot bit-flip enable per position in the beat
out_root_count	out	$\lceil \log_2(T_BITS + 1) \rceil$	roots found so far
out_last	out	1	final position of the block

8.4 Microarchitecture internals

8.4.1 Parallel Horner cells

$\Lambda(x)$ is evaluated at consecutive powers of α using Horner cells. For position i , the evaluation is:

$$\Lambda(\alpha^{-i}) = \Lambda_0 + \Lambda_1 \alpha^{-i} + \Lambda_2 \alpha^{-2i} + \dots + \Lambda_t \alpha^{-ti}$$

Each cell holds a running partial sum and multiplies by a constant factor each cycle, stepping from one position to the next. With $\text{BITS_PER_BEAT} = B > 1$, each lane evaluates B positions per cycle by chaining B constant multiplies from the same starting register.

The root flag per position is:

$$w_root[i] = (w_lambda_at_pos[i] == 0)$$

8.4.2 Direct bit-flip accumulation

At this fub level $out_flip_en[i]$ is the raw root flag for position i (gated only by position validity in the final partial beat). The qualifier $w_root[i] \ \&\& \ w_correctable \ \&\& \ !$ $w_release_passthrough$ lives in $bch_decoder_core$, which owns $w_correctable$ (root count matches locator degree, degree within T_BITS) and $w_release_passthrough$ (uncorrectable and frame-err blocks pass through unchanged, R2). There is no error-value computation anywhere; every root means flip.

8.4.3 Uncorrectability check

The block tracks the root count across the N_BITS positions:

$$w_correctable = (r_root_count == r_lambda_degree) \ \&\& \ (r_lambda_degree \leq T_BITS)$$

If the root count differs from the degree, or if the degree exceeds T_BITS , the block is uncorrectable. The final $w_correctable$ value is captured on the last position and passed to the decoder core as the block verdict.

8.5 FSM policy

This block carries **no FSM** (per vault/handbook/design/streaming-no-fsm.md). The Chien walk is driven by:

- `r_position`: counts bit positions, 0 .. `N_BITS`-1
- `r_root_count`: running count of roots
- `out_valid`: registered pipeline flag
- `w_last`: combinational qualifier for the final position

The position counter replaces any state machine.

8.6 Timing

- Serial case ($\text{BITS_PER_BEAT} = 1$): N_BITS cycles per block to evaluate all positions.
- Parallel-beat case: $\text{ceil}(N_BITS / \text{BITS_PER_BEAT})$ cycles per block, with B Horner evaluations per lane per cycle.
- The exact cycle count and throughput trade-off are placeholders tied to PRD D6.

8.7 Notes

- A shortened code ($N_BITS < 2^m - 1$) is handled by starting the Chien walk at the appropriate offset so that the first emitted position corresponds to bit 0 of the shortened codeword.
- The no-Forney structure is a key formal target: prove that every root of Λ in a correctable block corresponds to exactly one bit flip and that the resulting corrected block has zero syndrome.
- `out_flip_en` is a beat-wide vector; the corrector XORs it with the corresponding beat from the block buffer.

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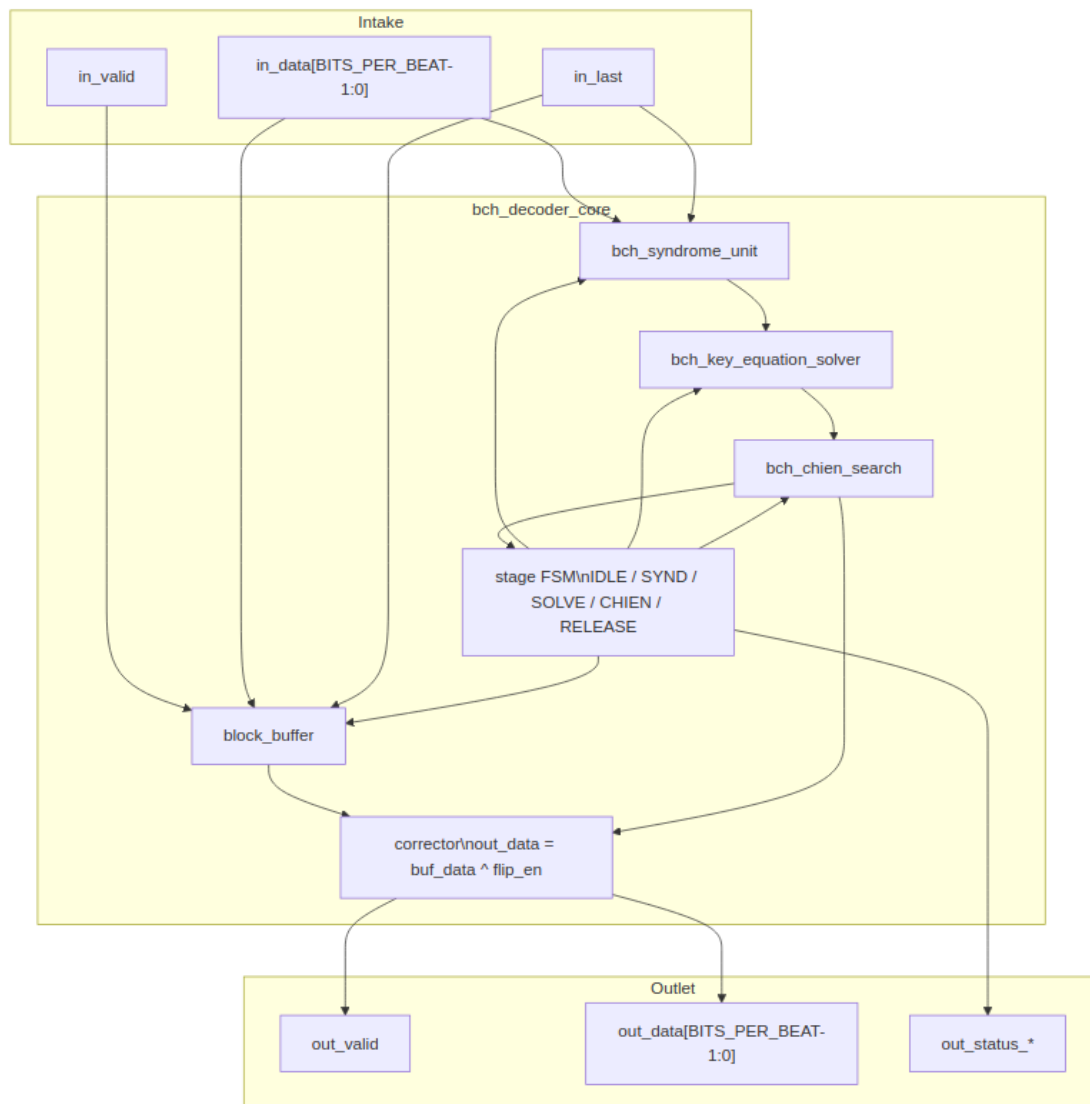
9 Decoder Core Integration (bch_decoder_core)

Module: `bch_decoder_core.sv` **Location:** `rtl/macro/` **Category:** integration block + one minimal control FSM **Parent:** `bch_top / standalone decoder wrapper` **Status:** landed — `rtl/macro/bch_decoder_core.sv`. The sequencer has a sixth state RECHECK (IDLE/SYND/SOLVE/CHIEN/RECHECK/RELEASE): the verdict is computed only after the optional re-check syndrome unit finishes, because the re-check result is not known until the corrected stream's last beat. Corrections are applied at the output read, not during the Chien walk (release-on-verdict, mirroring `rs_decoder_core`), so the MAS timing note “first corrected beat leaves as soon as the Chien walk starts” is corrected: the corrected image is computed during the walk but released only after the verdict. Single outstanding block; pipelining is PRD D6 open work. Known limitation: a block longer than `N_BITS` deadlocks the input (named in the RTL header).

9.1 Purpose

bch_decoder_core integrates the received-block buffer, syndrome unit, key-equation solver, Chien search, and corrector into one valid/ready streaming block. It is the only block in the decoder datapath that is allowed to contain a control FSM, because the sequencing of distinct algorithmic stages is genuine control, not per-beat datapath.

9.1.1 Figure 2.4: Decoder core block diagram



Decoder core block diagram

9.2 Parameters

Table 2.10: Decoder core parameters

Parameter	Type	Range	Default	Meaning	PRD
FIELD_DIM	int	3..16	TBD	m; field is $GF(2^m)$	D1
T_BITS	int	$1..(2^m-1)/2$	TBD	correctable bit errors	D2
N_BITS	int	$2t+1 .. 2^m-1$	TBD	codeword length in bits	D3
BITS_PER_BEAT	int	1 .. K_BITS	TBD	bits per valid/ready beat	D9
ENABLE_ERASURES	bit		0 TBD	erasure input and erasure-locator logic	D5
KES_ALGO	string	"RIBM", "EUCLID", "SMALL_T"	TBD	solver algorithm	D11

9.3 Interface

Table 2.11: Decoder core ports

Signal	Direction	Width	Description
aclk	in	1	one clock
aresetn	in	1	active-low synchronous reset
in_valid / in_ready	in / out	1	valid/ready for received bits
in_data	in	BITS_PER_BEAT	received bits
in_keep	in	BITS_PER_BEAT	present-bit mask
in_last	in	1	this beat carries the N_BITS-th received bit
out_valid / out_ready	out / in	1	valid/ready for corrected data
out_data	out	BITS_PER_BEAT	corrected data bits; parity lanes dropped
out_keep	out	BITS_PER_BEAT	present-bit mask
out_last	out	1	this beat carries the K_BITS-th data bit
out_status_ok	out	1	block had no errors
out_status_corre cted	out	$\lceil \log_2(T_BITS + 1) \rceil$	bits corrected (0 .. T_BITS)
out_status_uncor rectable	out	1	correction failed
out_status_frame _err	out	1	block length was not N_BITS

9.4 Microarchitecture internals

9.4.1 Block buffer

The received block is written into a buffer as it arrives and read out during the Chien/correction pass. The buffer depth is at least `N_BITS` bits, sized to absorb the latency of the syndrome pass and the solver. For an uncorrectable block, the buffer passes the received bits through unchanged (R2):

```
w_release_passthrough = w_uncorrectable || w_frame_err
```

9.4.2 Stage sequencer

The decoder core contains exactly one minimal control FSM (per vault/handbook/design/minimal-fsm.md). Stage sequencing is control, not per-beat datapath. The proposed state list merges any state that only waits one fixed cycle into its successor with a counter:

```
IDLE    -> wait for a new block to begin arriving
SYND     -> receiving the block into the buffer and the syndrome unit
SOLVE    -> key-equation solver running
CHIEN    -> Chien search / correction pass over the buffer
RELEASE  -> outputting the corrected (or passthrough) block
```

A state like “wait one cycle for solver start” is killed: it becomes a registered `r_solver_start` pulse inside SYND or SOLVE.

9.4.3 Corrected-stream re-check

As an optional R2 mechanism, a second syndrome unit may accumulate over the corrected stream leaving the buffer. If any odd syndrome is non-zero after correction, the block is flagged uncorrectable even if the Chien root count matched the locator degree. This mirrors the release-on-verdict discipline in the sibling `rs_decoder_core` architecture. The re-check is gated by a `build` parameter and is an implementation detail, not part of the locator search.

9.4.4 Correctability verdict

The per-block uncorrectable verdict is:

```
w_uncorrectable = !w_correctable || (w_recheck_enabled && !
w_recheck_zero)
w_correctable = (r_root_count == r_lambda_degree) && (r_lambda_degree
<= T_BITS)
```

`out_status_uncorrectable` is held for every beat of the released block.

9.5 FSM policy

This block is the **one place in the decoder datapath where a minimal control FSM is permitted**. The FSM has at most the four states above; any state that only waits one fixed cycle is merged with its successor. All per-beat operations inside the stages (syndrome accumulation, solver array stepping, Chien evaluation) remain FSM-free pipelines.

9.6 Timing

- SYND phase: $\text{ceil}(N_BITS / BITS_PER_BEAT)$ beats.
- SOLVE phase: depends on KES_ALGO and T_BITS (PRD D11).
- CHIEN / RELEASE phase: $\text{ceil}(N_BITS / BITS_PER_BEAT)$ beats, overlapped so that the first corrected beat leaves as soon as the Chien walk starts.
- Total latency and throughput trade-offs are placeholders tied to PRD D6.

9.7 Notes

- This whole page is conditional on the build including the decoder (PRD D4) and on the throughput architecture (PRD D6) and solver choice (PRD D11).
- Frame error: a block whose `in_last` arrives before or after the `N_BITS`-th bit is flagged with `out_status_frame_err` and passes through unchanged. The first `length - (N_BITS - K_BITS)` bits of a too-long block are treated as data.
- Status outputs describe the block being emitted and are valid on every beat of it; a consumer may sample them with `out_last`.

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10 Core Signals

This page presents the valid/ready core signals as a contract table rather than restating the prose interface description in `../bch_has/ch04_interfaces/01_core_interface.md`. The direction, reset value, and creator block for each signal are the source of truth here.

10.1 Reset strategy

All core logic uses **synchronous active-low reset** (`aresetn`), the house clocking convention. Reset clears block state, counters, the LFSR, syndrome registers, solver arrays, and the Chien position counter. The block buffer itself is not reset; its read/write pointers are reset, which makes the buffer contents logically invalid until overwritten.

10.2 bch_encoder_core signal contract

Table 3.1: bch_encoder_core signal contract

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
aclk	in	1	—	clock source	one clock for the whole core	—
aresetn	in	1	—	reset source	synchronous active-low reset	—
in_valid	in	1	—	upstream	stable until in_ready high	encoder core
in_ready	out	1	0	encoder core	combinational ! r_valid m_ready gated by parity-drain	encoder core
in_data	in	BITS_PER_BEAT	—	upstream	data bits, low-aligned; partial final beat qualified by in_keep	encoder core
in_keep	in	BITS_PER_BEAT	—	upstream	present-bit mask, low-aligned;	encoder core

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
in_last	in	1	—	upstream	partial only on final beat this beat carries the K_BITS-th data bit	encoder core
out_valid	out	1	0	encoder core	registered pipeline flag; data beats then parity beats	encoder core
out_ready	in	1	—	downstream	may depend on out_valid	encoder core
out_data	out	BITS_PER_BEAT	0	encoder core	data while ! w_parity_phase, parity while w_parity_phase	encoder core
out_keep	out	BITS_PER_BEAT	0	encoder core	full except on final data and final parity beats	encoder core
out_last	out	1	0	encoder	this beat	encoder

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
				core	carries the N_BITS-th coded bit	core
frame_err	out	1	0	encoder core	pulse when in_last does not coincide with K_BITS data bits	encoder core

10.3 bch_decoder_core signal contract

Table 3.2: bch_decoder_core signal contract

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
in_valid	in	1	—	upstream	stable until in_ready high	decoder core
in_ready	out	1	0	decoder core	high while buffer has space; drops when full	decoder core
in_data	in	BITS_PER_BEAT	—	upstream	received bits, low-aligned	decoder core
in_keep	in	BITS_PER_BEAT	—	upstream	present-bit mask; partial only on final beat	decoder core
in_last	in	1	—	upstream	this beat carries the N_BITS-th received bit	decoder core
in_erase	in	BITS_PER_BEAT	—	upstream	(only with ENABLE_ERASURES) known-bad bits	decoder core

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
out_valid	out	1	0	decoder core	high once the block verdict is final	decoder core
out_ready	in	1	—	downstream	may depend on out_valid	decoder core
out_data	out	BITS_PER_BEAT	0	decoder core	corrected data bits; parity lanes dropped	decoder core
out_keep	out	BITS_PER_BEAT	0	decoder core	present-bit mask; partial only on final beat	decoder core
out_last	out	1	0	decoder core	this beat carries the K_BITS -th data bit	decoder core
out_status_ok	out	1	0	decoder core	held for every beat: block had no errors	syndrome unit, decoder core
out_status_corrected	out	$\lceil \log_2(T_BITS+1) \rceil$	0	decoder core	held for every beat: bits corrected (0 .. T_BITS)	Chien search, decoder core

Signal	Dir	Width	Reset value	Creates / consumes	Contract	Source block page
out_stat us_uncor rectable	out	1	0	decoder core	held for every beat: correction failed; data pass through unchanged	Chien search, decoder core
out_stat us_frame _err	out	1	0	decoder core	held for every beat: block length was not N_BITS	decoder core

10.4 Timing rules

- `in_last` is the sole block boundary; the cores do not infer boundaries by counting.
- `out_status_*` describe the block being emitted and are valid on every beat of it, first to last. A consumer may sample them with `out_last` or with the first beat.
- An uncorrectable block leaves exactly as received; the corrector is disabled by `w_release_passthrough`.

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11 Signal Contracts and K-maps

11.1 What a contract is

A signal contract in this repo is a machine-checkable statement of what a signal is allowed to do. Each contract has three parts, in order:

1. **Term list:** every signal or expression in play, with its defining expression and `file:line` citation.
2. **Invariants:** the strict relationships between terms, each cited, each stating which rows of the decision table it renders impossible.
3. **Decision table:** one row per combination of the terms. Rows the invariants forbid are marked **ILLEGAL**; legal rows carry the resulting output.

The canonical methodology is in `vault/handbook/design/signal-contracts-and-kmaps.md`; the shared generator machinery is in `bin/kmaps/`.

11.2 Which signals have contracts

The BCH workbook documents the control/qualifier signals whose correctness is non-obvious: parity-phase select, block-boundary and frame-error qualifiers, syndrome completion, no-error bypass, correctability verdict, Chien root-position enables, and uncorrectable pass-through. The complete list is in the generated workbook.

11.3 Pre-RTL citation posture

At MAS v0.1 there is no RTL, so the contracts cite the MAS pages where the intended expressions are written verbatim. When the first RTL lands, every MAS citation is re-pointed at the corresponding .sv file and line. The generator's citation gate then enforces that the workbook and the RTL agree; a drift fails the run.

11.4 Regenerating the workbook

```
python3  
projects/components/ecc-ip/bch/docs/gen_bch_signal_contracts_kmaps.py
```

The generator writes:

```
projects/components/ecc-ip/bch/docs/bch_signal_contracts.xlsx
```

Do not hand-edit the `.xlsx`; it is a build artifact of the generator.

11.5 Workbook contents

The generated workbook contains:

- a contract sheet for the core signals
- a kmap sheet per block with the key combinational decisions
- term lists, invariants, decision tables, `depends_only_on` sufficiency arguments, and `rtl_sop` expressions for pre-RTL review

For every kmap the verdict is currently NOT CHECKED because no RTL exists yet. That is the correct pre-RTL posture; the verdict becomes meaningful once `rtl_sop` can be diffed against the derived minimal cover.